The Lengths of Discrimination we go to in Maths

A Lineage of Measurement, from the Field Norm to the Inner Product

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On distance etymology

The word *discriminant* comes from the Latin *discriminare*, to divide, to separate, to distinguish one thing from another, and the mathematical object honours that root with unusual precision. The discriminant of the quadratic $ax^2 + bx + c$, $b^2 - 4ac$, is the quantity that decides whether the two roots stay together as a repeated value, separate into two real numbers, or fly apart into a conjugate complex pair. Similarly we will take a family of words (associated to length) that everyone treats as roughly interchangeable measuring instruments, asking what distinguishes them.

The family is this: *norm*, *quadratic form*, *bilinear form*, *inner product*, *metric*, and the discriminant itself, together with a few outriders such as the valuation and the height. Every one of these is, in some register, a way of assigning a size or a comparison to elements of a space, and every one of them is used with slightly different force in analytic number theory, in algebraic number theory, and in geometry. The words share ancestry, they overlap in meaning, and yet they are genuinely distinct objects, and the interesting question is how a mathematical community with no central authority came to agree on which word carries which meaning, a question we return to at the close.

Norm: the carpenter's square with two children

Norm descends from the Latin *norma*, a carpenter's set square, the physical tool used to check that an angle is true. From the tool came the sense of a standard, a rule against which things are measured, and from that sense mathematics took two quite separate objects that both deserve the name because both are standards of measurement.

The geometric norm on a vector space, written $\|v\|$, measures length. It is non-negative, it scales absolutely under multiplication by a scalar so that $\|\lambda v\| = |\lambda| \, \|v\|$, and it satisfies the triangle inequality $\|u + v\| \le \|u\| + \|v\|$. The number-theoretic norm $N(\alpha)$

of an algebraic number is a different creature: it is the product of the Galois conjugates of α, or equivalently the *determinant* of the map "multiply by α" acting on the field as a vector space over $\mathbb{Q}$. For a Gaussian integer $a+bi$ this determinant works out to a^2+b^2, which happens to equal the square of the Euclidean length, and that coincidence is the source of a great deal of confusion, because it invites the belief that the geometric and number theoretic norms are one idea. Rather they are separated by a single decisive property: the field norm is multiplicative, $N(\alpha\beta) = N(\alpha)N(\beta)$, a fact the geometric length never satisfies, and in real quadratic fields the field norm can be negative, as with $N(a+b\sqrt{2}) = a^2 - 2b^2$, which no length can ever be.

The agreement in $\mathbb{Z}[i]$ has a structural cause worth naming. In an imaginary quadratic field, multiplication by α is a rotation-and-scaling of the plane, so it scales area by $|\alpha|^2$, and the determinant reading of the norm therefore coincides with the squared-length reading. The Eisenstein integers $\mathbb{Z}[\omega]$ behave the same way, with norm $a^2 - ab + b^2$ equal to the squared length in the hexagonal lattice. The moment we pass to a real quadratic field the multiplication is a squeeze rather than a rotation, area is signed, and the norm inherits the sign, which is the technical reason Pell's equation and the units of $\mathbb{Z}[\sqrt{2}]$ become interesting.

```
# The field norm splits from length exactly when d > 0.
# N(a + b*sqrt(d)) = a^2 - d*b^2.

def field_norm(a, b, d):
    return a*a - d*b*b

print(field_norm(1, 1, -1))    #  2  : squared length of 1 + i
print(field_norm(1, 1,  2))    # -1  : 1 + sqrt(2) is a unit
```

That value of -1 marks $1+\sqrt{2}$ as the fundamental unit of $\mathbb{Z}[\sqrt{2}]$, and its powers generate every solution of $x^2-2y^2 = \pm 1$. The failure of the norm to be a length is the very thing that makes the arithmetic rich.

Quadratic form: the eldest of the line

Quadratic comes from *quadratus*, squared, from *quattuor*, four, by way of the four-sided figure. The quadratic form is the oldest object in our family, a homogeneous degree-two polynomial $Q(x) = \sum a_{ij}x_ix_j$, and its history runs through Fermat's characterisation of primes as $x^2 + y^2$, Lagrange's four-square theorem, and Gauss's composition of binary quadratic forms in the *Disquisitiones*. The field norm of a quadratic extension is literally a *binary quadratic form*, which is why the theory of forms and the theory of quadratic fields are two dialects describing one landscape, with Gauss's form class group foreshadowing the ideal class group by several decades.

Bilinear form: the comparison instrument

Bilinear joins *bi* to *linearis*, of a line, from *linea*, a linen thread stretched taut. A bilinear form $B(x, y)$ takes two vectors and returns a scalar, linearly in each argument, and it is the instrument that compares two vectors rather than measuring one. Over any field where dividing by two is allowed, the bilinear form and the quadratic form carry identical information, converted back and forth by the *polarization* identity,

$$B(x, y) = \tfrac{1}{2}\big(Q(x + y) - Q(x) - Q(y)\big).$$

The quadratic form is the diagonal restriction $Q(x) = B(x, x)$, and the bilinear form is its off-diagonal completion. In characteristic two the halving is unavailable and the two objects come apart, which is the kind of subtlety that matters in coding theory and in the arithmetic of function fields.

Inner product: bilinear form made positive and symmetric

The *inner product* sits inside the bilinear family as a special member, carrying its own history. The name comes from Grassmann and Gibbs in the nineteenth century, contrasting the "inner" product that returns a scalar with the "outer" product that returns a higher object, the inner and outer marking whether the multiplication draws the operands together into a number or throws them apart into a matrix or a form. An inner product $\langle x, y\rangle$ is a bilinear form that is symmetric and positive-definite, meaning $\langle x, x\rangle > 0$ for every non-zero x, and this positivity is what allows it to give birth to a genuine length.

The length lineage is now visible as a chain of specialisations, and it can be read in either direction, which is the conceptual hinge of this whole essay. Read downward in the constructive direction, we start from the richest object and build outward: an inner product $\langle x, y\rangle$, fed a vector against itself and square-rooted, $\|v\| = \sqrt{\langle v, v\rangle}$, yields a norm, and the norm of a difference, $d(x, y) = \|x - y\|$, yields a metric. On this reading the metric is the *most* structured object, the finished instrument, because it carries the inner product inside it as its engine, and this is the working geometer's and the statistician's view. Read upward in the axiomatic direction, we start from the metric as a bare list of three axioms and ask what extra properties would let us climb back: a metric that comes from a norm, a norm that comes from an inner product. On this reading the metric is the *least* structured object, the permissive floor. Both readings are correct, and the tension between them is exactly why students disagree about whether a metric is the humble base or the crowning construction. Each arrow runs one way only, because a general norm need not come from any inner product, the test being whether it obeys the parallelogram law, and a general metric need not come from any norm.

A test for where a given norm sits in this chain appears below once we have the metric in hand, since the cleanest way to feel the one-directional arrows is to catch a norm failing to be an inner product.

Metric: the permissive floor and the crowning construction

Metric comes from the Greek *metron*, measure, the same root standing inside *geometry*, earth-measurement. A metric $d(x, y)$ asks, in its bare axiomatic form, for very little: non-negativity, symmetry, the triangle inequality, and zero distance only between identical points. Read this way it carries no algebraic structure at all, no way to add points, no way to scale them, which is why the axiomatic direction places it at the permissive floor of the lineage. Read the other way, a metric built as $d(x, y) = \sqrt{\langle x - y, x - y\rangle}$ from a chosen inner product is the most structured object we can write, because the choice of inner product encodes an entire geometry, and this constructive reading is the one that the Mahalanobis distance below makes vivid.

With the metric now in view, here is the promised test of where a norm sits, using the parallelogram law, which holds exactly when a norm descends from an inner product.

```python
import numpy as np

# The parallelogram law decides whether a norm is inner-product-born.
# 2||x||^2 + 2||y||^2  ==  ||x+y||^2 + ||x-y||^2  holds iff it is.

def parallelogram_defect(x, y, p):
    n = lambda v: np.sum(np.abs(v)**p)**(1/p)   # the p-norm
    lhs = 2*n(x)**2 + 2*n(y)**2
    rhs = n(x+y)**2 + n(x-y)**2
    return lhs - rhs

x, y = np.array([1.0, 0.0]), np.array([0.0, 1.0])
print(round(parallelogram_defect(x, y, 2), 6))   # 0.0  : Euclidean, from inner product
print(round(parallelogram_defect(x, y, 1), 6))   # nonzero : taxicab, no inner product
```

Cursory reading of this snippet: the $p = 2$ norm returns a defect of zero and is therefore secretly an inner product wearing a length's clothing, while the $p = 1$ taxicab norm returns a non-zero defect and can never be one, so it lives at the metric floor with no inner product beneath it.

The ultrametric surprise

The metric is where analytic number theory rejoins the story with a jolt. The p-adic absolute value on $\mathbb{Q}$ measures the size of a rational by how divisible it is by a fixed prime p, and the metric it induces satisfies a strengthened triangle inequality, the ultrametric law $d(x,z) \le \max\bigl(d(x,y),\, d(y,z)\bigr)$. In this geometry every triangle is isosceles, and every point inside a ball serves as its centre, a world genuinely alien to Euclidean intuition. Ostrowski's theorem then delivers the punchline of the whole essay: up to equivalence, the only absolute values on $\mathbb{Q}$ are the ordinary Archimedean one and the p-adic ones, a single value for each prime. The primes themselves index the ways of measuring size on the rationals, and the abstract question "what are the possible metrics here" turns out to have the same answer as "what are the primes," which is the deepest junction between measurement and arithmetic that these words conceal.

```python
from fractions import Fraction

# p-adic valuation, then the p-adic absolute value |x|_p = p^{-v_p(x)}.
def v_p(x, p):
    x = Fraction(x)
    if x == 0:
        return float('inf')
    v, n, d = 0, x.numerator, x.denominator
    while n % p == 0: n //= p; v += 1
    while d % p == 0: d //= p; v -= 1
    return v

def abs_p(x, p):
    v = v_p(x, p)
    return 0.0 if v == float('inf') else p**(-v)

# Highly divisible by 2 means small in the 2-adic sense.
print(abs_p(8, 2))      # 0.125 : 8 is 2-adically small
print(abs_p(3, 2))      # 1.0   : 3 is a 2-adic unit
```

Note in passing how the intuition inverts: a number heavily divisible by p is p-adically small, so 8 is closer to 0 than 3 is when the prime is two, and this inversion is the entire reason the ultrametric geometry looks so strange.

Mahalanobis: the metric as the crowning construction

The constructive reading of the lineage, the one that places the metric at the top rather than the bottom, has a beautiful historical witness. In 1936 the Indian statistician P. C. Mahalanobis published *On the Generalized Distance in Statistics*, and he did something that ought to be far better known: he wrote down a distance between two probability distributions and noticed that its structure is identical to the line element of general relativity.[1] His generalised distance between two mean vectors μ_1 and μ_2 with shared covariance Σ is

$$\Delta \;=\; \sqrt{(\mu_1 - \mu_2)^\top \, \Sigma^{-1} \, (\mu_1 - \mu_2)}\,,$$

and Mahalanobis observed that his statistical line element $P \cdot \Delta^2 = \alpha_{\mu\nu}\,(d\alpha)^\mu (d\alpha)^\nu$ is the exact analogue of the relativistic $ds^2 = g_{\mu\nu}\, dx^\mu dx^\nu$, with the inverse covariance Σ^{-1} playing the part of the metric tensor $g_{\mu\nu}$.

Placed in our lineage, the Mahalanobis distance is a metric that visibly carries an inner product as its engine, since it is the norm induced by $\langle x, y\rangle_\Sigma = x^\top \Sigma^{-1} y$. When Σ is the identity the whole thing collapses to ordinary Euclidean length, which is Mahalanobis's "dispersion the same everywhere" case, the flat statistical field corresponding to special relativity. When Σ carries variance and correlation, the inner product bends, and distances are measured in units of standard deviation along each direction, so a gap of three along a high-variance axis counts for less than the same gap along a tight one.

```python
import numpy as np

# Mahalanobis distance = norm from the inner product <x,y> = x ^T Sigma_inv y.
# Sigma = I recovers Euclidean length (the "flat" statistical field).

def mahalanobis(mu1, mu2, Sigma):
    d = np.asarray(mu1) - np.asarray(mu2)
    Sigma_inv = np.linalg.inv(Sigma)
    return np.sqrt(d @ Sigma_inv @ d)          # sqrt of d^T Sigma^{-1} d

mu1, mu2 = [0, 0], [3, 3]

print(mahalanobis(mu1, mu2, np.eye(2)))                  # 4.243 :
```

[1] The volume in which this appeared, an English edition of the Einstein and Minkowski relativity papers, was translated by M. N. Saha and S. N. Bose, the same Saha of the Saha ionisation equation and the same Bose of Bose–Einstein statistics, so the differential-geometric line element and its statistical twin were circulating through one small circle of Calcutta physicists at once.

```
    Euclidean, Sigma = I
print(mahalanobis(mu1, mu2, [[4, 0], [0, 1]]))        # 3.354 :
    x-axis stretched, x-gaps count less
print(mahalanobis(mu1, mu2, [[1, 0.9], [0.9, 1]]))    # 3.078 :
    positive correlation, gap runs along the grain
```

The covariance matrix Σ is the metric tensor of the statistical space, and inverting it is what turns "spread of the data" into "cost of distance," so directions the data explores freely become cheap to travel and directions it resists become expensive. This is the metric at its most structured, the crowning construction of the constructive reading.

The discriminant returns, and its outriders

Having opened on the discriminant we can now place it exactly. The discriminant of a quadratic form, or of a number field, is computable as the determinant of the bilinear pairing on a chosen basis, and it measures how far the lattice of integers departs from being a perfect square grid. It distinguishes, once again, this time telling us where a field ramifies, which primes behave badly, and whether a form is degenerate. Two further outriders round out the map. The *valuation*, from *valere*, to be strong or worthy, is the logarithmic shadow of the p-adic absolute value, converting multiplicative questions into additive ones. The *height* of a rational point, from the ordinary sense of vertical extent, measures arithmetic complexity rather than geometric size, so that $H(p/q) = \max(|p|, |q|)$ assigns very different values to $1/2$ and $999/1998$ even though a metric would call them one and the same.

Capstone: the king on a shrinking board

Every object in the lineage can be brought to bear on a single concrete animal, the chess king, and doing so reveals a divergence that the abstract definitions only hint at. The king is the ideal specimen because his native way of measuring distance is one specific member of the norm family we have been discussing, and because the board he moves on is a graph whose adjacency matrix changes as he walks, which lets us watch a single-vector length and a whole-network quadratic form come apart.

The king's own metric is Chebyshev

A king crosses one square per move whether he steps orthogonally or diagonally, so the number of moves he needs between two squares is the Chebyshev distance,

$$d_\infty(u, v) = \max\big(|\Delta r|, \, |\Delta c|\big),$$

which is the $p \to \infty$ limit of the p-norm family, the far extreme from the $p = 1$ taxicab norm we met when the parallelogram law failed. This is why a king reaches the opposite corner of an empty board in seven moves rather than the fourteen an orthogonal-only piece would need, since under L^∞ the diagonal step costs exactly the same as a straight one. The three headline norms sit on the same displacement vector and differ only in how they weigh its components.

```
import numpy as np

# Three norms of the same corner-to-corner displacement (7, 7).
d = np.array([7, 7])
print(int(np.max(np.abs(d))))        # 7     : Chebyshev (L-inf), the king's move count
print(int(np.sum(np.abs(d))))        # 14    : taxicab   (L1), orthogonal steps only
print(round(np.sqrt(d @ d), 3))      # 9.899 : Euclidean (L2), the straight ruler
```

Cursory reading: the same vector $(7, 7)$ returns three different sizes, all obeying the norm axioms, but only the L^2 value descends from an inner product and so passes the parallelogram law, which is the one-directional arrow the essay established earlier; the king simply happens to live at L^∞.

The board is a graph, and Chebyshev is its geodesic

On the full board the king's Chebyshev distance is exactly the shortest-path distance in the king's-move graph, where each square is a node joined to its up to eight neighbours. The degree of a node is the count of legal moves available from it, which falls at the boundary: a corner square offers three moves, an edge square five, an interior square eight. This degree is the diagonal of the degree matrix D, and the adjacency matrix A records the moves themselves, so the two matrices between them hold the entire combinatorial structure the king navigates.

Now the crucial step. We cannot feed the *adjacency matrix* A straight into the Mahalanobis quadratic form, because A is symmetric but has a zero diagonal and eigenvalues of both signs, so it is not positive-definite and A^{-1} would produce negative squared lengths, failing every metric axiom. The correct positive-semidefinite stand-in for the covariance is the graph Laplacian $L = D - A$, whose pseudo-inverse L^+ slots into exactly the place Σ^{-1} occupied in the Mahalanobis distance, giving the effective resistance distance,

$$R(u, v) = \sqrt{(e_u - e_v)^\top L^+ (e_u - e_v)}\,,$$

which is a genuine metric and is the honest network analogue of Mahalanobis, the same quadratic form eating an indicator difference instead of a mean difference.

Where the two lengths diverge

On the empty board the geodesic length and the resistance length tell compatible stories, but they measure fundamentally different things, and the difference surfaces the instant the board shrinks. The Chebyshev geodesic counts only the single cheapest chain of moves and is blind to how many alternative routes exist, whereas the resistance distance feels the entire remaining lattice and rises whenever parallel routes are removed, because electrically, fewer parallel paths mean higher resistance.

```
import numpy as np, networkx as nx

def king_graph(n):
    G = nx.Graph()
    for r in range(n):
        for c in range(n):
            for dr in (-1, 0, 1):
                for dc in (-1, 0, 1):
                    if (dr or dc) and 0 <= r+dr < n and 0 <= c
                        +dc < n:
                         G.add_edge((r, c), (r+dr, c+dc))
    return G

def resistance(G, u, v):                        # Mahalanobis-shaped:
    L^+ as Sigma^{-1}
    nodes = list(G.nodes())
    Lplus = np.linalg.pinv(nx.laplacian_matrix(G, nodelist=
        nodes).toarray().astype(float))
    e = np.zeros(len(nodes)); e[nodes.index(u)], e[nodes.index
        (v)] = 1, -1
    return np.sqrt(e @ Lplus @ e)

G = king_graph(8)
u, v = (0, 0), (7, 7)
print(nx.shortest_path_length(G, u, v))          # 7      :
    Chebyshev geodesic
print(round(resistance(G, u, v), 4))             # 1.1267 :
    whole-network length

for sq in [(0,7),(1,6),(7,0),(6,1),(2,5),(5,2)]: # carve the
    anti-diagonal region
    if sq in G: G.remove_node(sq)
```

```
print(nx.shortest_path_length(G, u, v))          # 7      : geodesic UNMOVED
print(round(resistance(G, u, v), 4))             # 1.1470 : resistance HAS RISEN
```

Cursory reading: removing squares that lie off the cheapest diagonal chain leaves the Chebyshev geodesic stubbornly at seven, because that one path survives intact, while the resistance distance climbs from 1.1267 to 1.1470, because the deleted squares were carrying parallel current that no longer flows; the gap between the two numbers is precisely the shrinking of possibilities that a plain geodesic cannot see. This is the payoff of the whole lineage: a metric records the length of one best route, whereas the Mahalanobis-shaped quadratic form, with its positive-semidefinite tensor in the middle, records the connectivity of the entire space, and the king's dwindling board is the cleanest place to feel the difference. The King's journey is a Chebyshev length when we ask how far, and a resistance length when we ask how freely, and the two answers separate exactly as the board closes in around him.

Convergent Solutions, before there were machines to check?

The striking thing about this length family is that a scattered community of mathematicians, working across centuries, in Latin and French and German and Italian and English, with no committee and no enforcement, arrived at a terminology that is mostly consistent, so that a norm means nearly the same thing to an algebraist in Bonn and an analyst in Kyoto. It is worth asking how that convergence happened, since it happened entirely without the large statistical language models that now flatten usage toward a mean.

Several forces did the work that no authority imposed. There was the gravity of a few canonical texts, Gauss's *Disquisitiones*, Dedekind's supplements to Dirichlet, Bourbaki's relentless systematisation, each of which fixed a vocabulary that later writers found it easier to adopt than to fight. There was the discipline of translation, since a term that could not survive being carried between languages tended to be quietly replaced by one that could, and the Latin and Greek roots served as a shared substrate precisely because no living nation owned them. There was the harsh filter of the referee and the seminar, where an idiosyncratic coinage met immediate resistance and a well-chosen one spread by imitation, a slow peer-to-peer process closer to how a spoken dialect settles than to how a standard is decreed. And there was the deepest constraint of all, that the mathematics itself pushed back: a word attached to an object with clean properties, multiplicativity for the norm, positivity for the inner product, would stick, while a word attached to something ill-defined would slide off, so the structures were doing much of the standardising on the community's behalf.

The convergence was therefore never total, which is the honest part of the story. The word "norm" still means two different things depending on whether you are measuring a vector or an algebraic number, and only context saves us, and "metric" means one thing to a topologist and a subtly enriched thing to a Riemannian geometer. What the community achieved was not a dictionary but a working consensus, robust enough for a paper written in one subfield to be read in another, loose enough to let each subfield keep the shadings it needed. That is a remarkable piece of distributed coordination, achieved by reading each other carefully and by letting the objects themselves adjudicate, and it is worth holding onto as a model of how a shared language can emerge from many hands attending to the same reality, without any single voice, human or otherwise, being given the final say.

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